

Import File

```
from google.colab import drive  
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import pandas as pd  
import numpy as np
```

```
df = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/imdb.csv")
df.head()
```

	web_scraper_order	web_scraper_start_url	title
0	1768461327-1	https://www.imdb.com/title/tt36642456/reviews/...	Simply impactful story telling
1	1768461327-2	https://www.imdb.com/title/tt36642456/reviews/...	Bold and Brilliant Film Inspired by Real Events
2	1768461327-3	https://www.imdb.com/title/tt36642456/reviews/...	Haq se dekho
3	1768461327-4	https://www.imdb.com/title/tt36642456/reviews/...	Great movie, Simple yet impactful
4	1768461327-5	https://www.imdb.com/title/tt36642456/reviews/...	A must watch movie

Install Transformer

```
!pip install -q transformers datasets sentence-transformers acceler
```

Create Small Sample

```
df = df.dropna(subset=['review'])
df["review"] = df["review"].astype(str)
df_demo = df.sample(n=20, random_state=42)
```

Sentiment Analysis using Transformers (cardiffnlp/twitter-roberta-base-sentiment)

```
label_map = {
    "LABEL_0": "Negative",
    "LABEL_1": "Neutral",
    "LABEL_2": "Positive",
}
```

```
from transformers import pipeline
sentiment = pipeline("sentiment-analysis", model="cardiffnlp/twitte
df_demo["sentiment_scores"] = df_demo["review"].apply(lambda x:{lab
for
df_demo["dominant_sentiment"] = df_demo["sentiment_scores"].apply(l
```

[Show hidden output](#)

df_demo

[Show hidden output](#)

Perform Emotion Analysis "j-hartmann/emotion-english-distilroberta-base"

```
emotion = pipeline("text-classification", model="j-hartmann/emotion
```

[Show hidden output](#)

```
df_demo["emotion_scores"] = (df_demo["review"].apply(lambda x: {
    item["label"]:item["score"]
    for item in emotion(x[:512])[0]
})))
```

df_demo

[Show hidden output](#)

Manual Classification of Text "valhalla/distilbart-mnli-12-1"

```
frames = [
    "gender justice",
    "religion reforms",
    "legal rights",
    "patriarchy and power",
    "artistic excellence",
    "social awareness",
    "emotional storytelling",
    "women empowerment"
]

frame_pipe = pipeline("zero-shot-classification",
                       model="valhalla/distilbart-mnli-12-1")

texts = df_demo["review"].str[:512].tolist()

results = frame_pipe(texts, candidate_labels=frames, batch_size=4)

df_demo["dominant_frame"] = [r["labels"][0] for r in results]
```

[Show hidden output](#)

df_demo

	web_scraper_order	web_scraper_start_url	text
68	1768461327-69	https://www.imdb.com/title/tt36642456/reviews/...	A Reflection of Str
45	1768461327-46	https://www.imdb.com/title/tt36642456/reviews/...	Bold, Beautiful Pathbreak
9	1768461327-10	https://www.imdb.com/title/tt36642456/reviews/...	HAQ r
76	1768461327-77	https://www.imdb.com/title/tt36642456/reviews/...	Truth in F
141	1768461327-142	https://www.imdb.com/title/tt36642456/reviews/...	Emotional Intel and D H
108	1768461327-109	https://www.imdb.com/title/tt36642456/reviews/...	An H Sl Hun

212	1768461327-213	https://www.imdb.com/title/tt36642456/reviews/...	Fan movie i
237	1768461327-238	https://www.imdb.com/title/tt36642456/reviews/...	Its a v
129	1768461327-130	https://www.imdb.com/title/tt36642456/reviews/...	A Story Needed
218	1768461327-219	https://www.imdb.com/title/tt36642456/reviews/...	Must Wa
87	1768461327-88	https://www.imdb.com/title/tt36642456/reviews/...	A Jo Th Emotio Co
33	1768461327-34	https://www.imdb.com/title/tt36642456/reviews/...	True ci craftsma
259	1768461327-260	https://www.imdb.com/title/tt36642456/reviews/...	A Pow Grou Drama Career
70	1768461327-71	https://www.imdb.com/title/tt36642456/reviews/...	Am
47	1768461327-48	https://www.imdb.com/title/tt36642456/reviews/...	Nic
112	1768461327-113	https://www.imdb.com/title/tt36642456/reviews/...	Unforge Sim
6	1768461327-7	https://www.imdb.com/title/tt36642456/reviews/...	Must V Mov
30	1768461327-31	https://www.imdb.com/title/tt36642456/reviews/...	A good time v
105	1768461327-106	https://www.imdb.com/title/tt36642456/reviews/...	Bril S D M
56	1768461327-57	https://www.imdb.com/title/tt36642456/reviews/...	Haq - A Watch of Stor

✓ Task

Analyze the sentiment and key discussion themes of the movie reviews

Visualize the distribution of dominant sentiments (Positive, Neutral, Negative) and dominant frames (e.g., gender justice, artistic excellence) found in the `df_demo` DataFrame to understand the overall public opinion and the most frequently discussed topics in the reviews.

✓ Analyze Dominant Sentiment Distribution

Subtask:

Calculate the distribution of 'dominant_sentiment' in the `df_demo` DataFrame to understand the overall sentiment of the reviews.

Reasoning: I will calculate the value counts of the 'dominant_sentiment' column in the `df_demo` DataFrame to get the sentiment distribution.

```
sentiment_distribution = df_demo['dominant_sentiment'].value_counts  
print(sentiment_distribution)
```

```
dominant_sentiment  
Positive      19  
Neutral       1  
Name: count, dtype: int64
```

Reasoning: The sentiment distribution has been calculated; now, I will visualize it using a bar plot to provide a clear graphical representation of the sentiment breakdown.

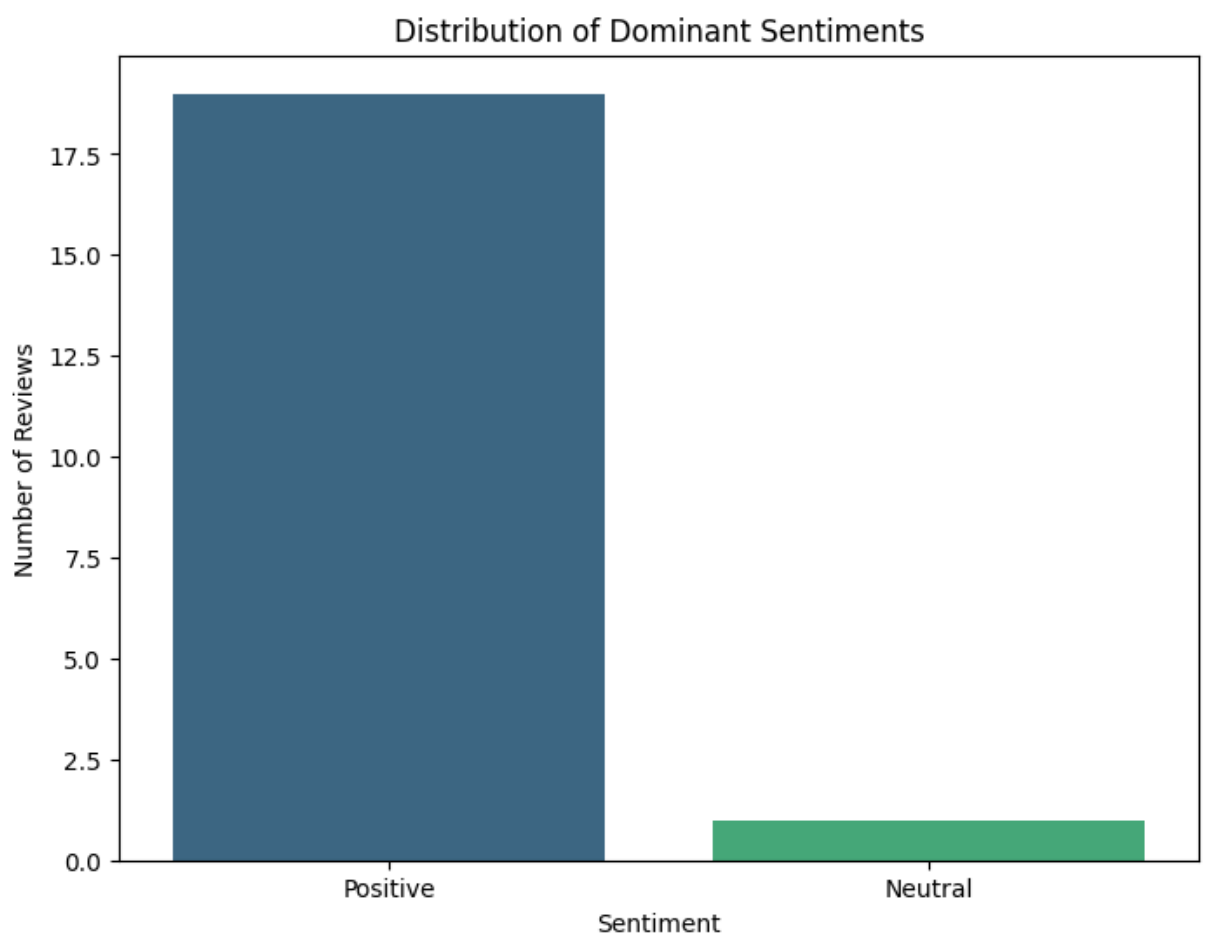
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 6))
sns.barplot(x=sentiment_distribution.index, y=sentiment_distributi
plt.title('Distribution of Dominant Sentiments')
plt.xlabel('Sentiment')
plt.ylabel('Number of Reviews')
plt.show()
```

/tmp/ipython-input-530330619.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be

```
sns.barplot(x=sentiment_distribution.index, y=sentiment_distributi
```



Reasoning: The previous code block produced a `FutureWarning` regarding the use of `palette` without `hue` in `seaborn.barplot`. To resolve this, I will explicitly set the `hue` argument to the x-axis variable and `legend=False` as suggested by the warning, ensuring cleaner code and preventing future deprecation issues.

Reasoning: I have successfully calculated and visualized the distribution of dominant sentiments. Now, to address the second part of the main task, I will calculate the distribution of 'dominant_frame' in the `df_demo` DataFrame to understand the most frequently discussed topics.

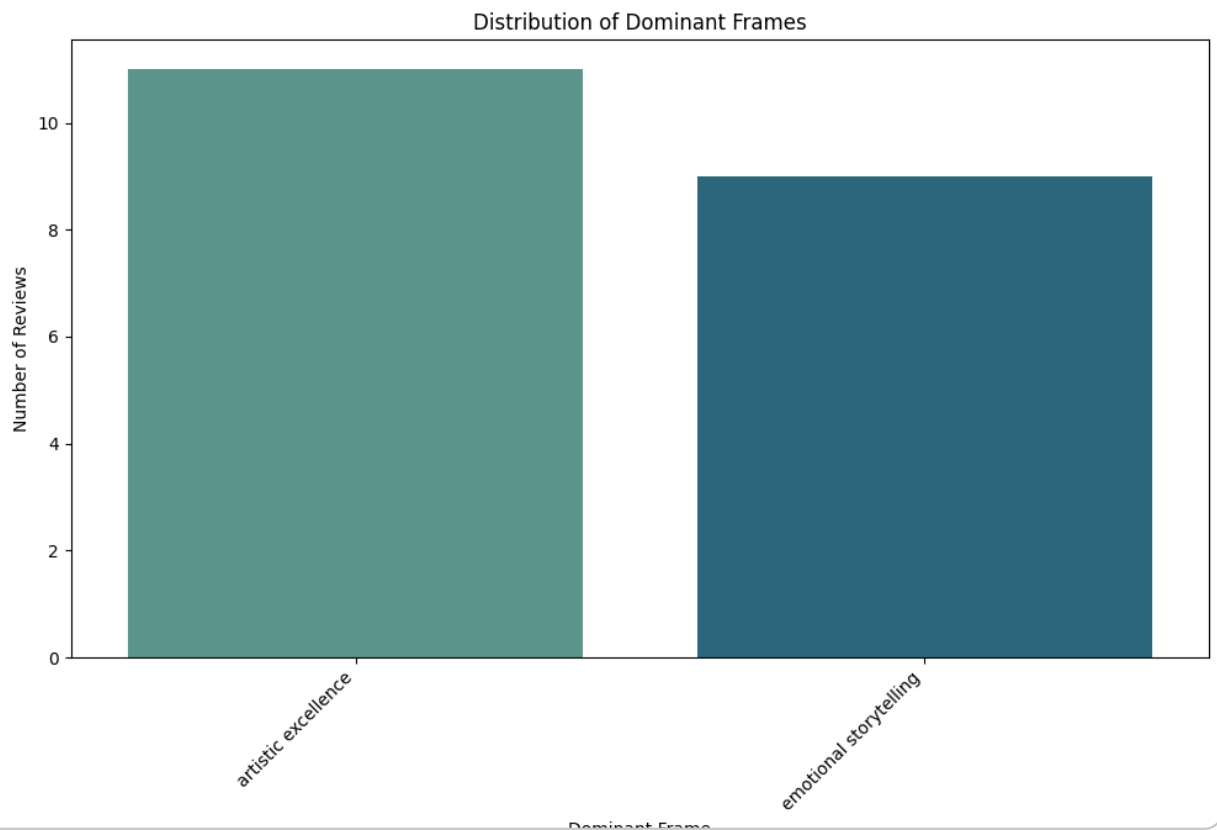
```
frame_distribution = df_demo['dominant_frame'].value_counts()
print(frame_distribution)
```

```
dominant_frame
artistic excellence    11
emotional storytelling    9
Name: count, dtype: int64
```

Reasoning: The distribution of dominant frames has been calculated. To visualize the most frequently discussed topics, I will now create a bar plot for the `frame_distribution`.


```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 7))
sns.barplot(x=frame_distribution.index, y=frame_distribution.values)
plt.title('Distribution of Dominant Frames')
plt.xlabel('Dominant Frame')
plt.ylabel('Number of Reviews')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Final Task

Subtask:

Summarize the findings from the sentiment and frame analysis, providing insights into public opinion and key discussion topics.

Summary:

Data Analysis Key Findings

- The majority of the movie reviews analyzed are positive, with 19 reviews classified as 'Positive' and only 1 as 'Neutral'. No negative sentiments were found in the dataset.
- The most frequently discussed themes (dominant frames) in the reviews are 'artistic excellence' with 11 instances, followed by 'emotional storytelling' with 9 instances.
- Both the distribution of dominant sentiments and dominant frames were successfully calculated and visualized using bar plots.

Insights or Next Steps

- The overwhelmingly positive sentiment suggests high satisfaction among viewers, indicating the movie is well-received.
- The prominence of 'artistic excellence' and 'emotional storytelling' as dominant frames highlights that these aspects are key drivers of audience engagement and critical appreciation. Future marketing could leverage these specific strengths.